

ActInf ArtStream 004.1 ~ Karin Kukkonen: "Literature as Uncertainty Practice"

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hello and welcome it's December 10th 2024 we're in active inference art stream number 4.1 with Karin kinan and federo and Bruno guests today we'll be discussing literature as uncertainty practice creative encounters with the world there will be a presentation followed by some comments and discussion so thank you all for joining and to you Ken for the presentation thank you thank you Daniel for inviting me um I'm speaking here from the Norwegian Institute in Rome um and I'm very much looking forward to the responses from Bruno and Federico and um whatever questions may come up uh in the chat later on I'm going to talk about predictive processing in relation to literature from the one side looking at ways in which we can use the framework of predictive processing in order to make sense of what literature is and what it does um but also going the other way thinking about what elements of predictive processing literature might um pose a challenge for or um provide a very interesting case study to perhaps provoke further discussion a lot of what I'm going to say is based on my book probability designs which came out with Oxford University press in 2020 um I'm going to introduce the concept um I'm going to relate it to the concept of form which is a concept from that is much used in literary studies um I'm going to raise the question of what makes literature different from other art forms and I'm going to link that to a very particular kind of cognitive work that literally form the long form written of um a novel enables um and then in a final move I'm going to talk about the ways in which literature works as an uncertainty practice in the world and what this has to do with um the general experience of contingency so let's start with the probability designs um and being a literary scholar I'll start with a textual example this is from Virginia W's novel Orlando um which is a novel where the main character starts as a man in the Renaissance then suddenly turns into a woman in the 18th century and carries on as a woman until um about the 20th century at this point in the book we're still in the Renaissance young Orlando has decided to become a poet until one day he looks out of the window and see he sees the splendid natural Green in the garden after that and here I quote

Virginia Woolf of course he could write no more green in nature is one thing Green in literature another nature and Ice seem to have a mutual antipathy bring them together and they tear each other to pieces the Shade of Green Orlando now saw spoiled his rhyme and split his meter so what we have here on a very basic level is of course a prediction error he wanted to become a poet um he looks outside of know he sees something surprising and he changes his plan in terms of a narrative such prediction errors would be plot events here something happens that changes fundamentally the trajectory um of the further course of the narrative Orlando leaves his desks and his plans of becoming a writer and what is likely to happen next in the novel has just changed these prediction errors are of course um modulated with Precision with our sense of how just how good that information is just how reliable it is and in literary text that is quite often tied to the character's perspective um so it's Orlando who sees that green is part of his perception um and it's very often also tied to the embodied experience of the characters and the way in which they're positioned in the fictional World in particular in the long form of the novel and this has been discussed in literary studies for a very long time at least since M B's idea that the novel is a polyphonic genre genre of many voices you usually have multiple different perspectives on the Character level but also on the level of the narrator who can be more or less reliable so that's very much um part of a standard uh setup for the novel to have these multiple different um perspectives but it's not just the spatial position um that gives um Precision in a literary text it is also on the level of temporality so if we take the last um sentence it has a particular Rhythm um that changes Midway through the sentence so the Shade of Green Orlando is I am then you have the now saw where the Rhythm sort of changes and that's of course exactly the moment of the prediction error spoils his rhyme and split his meter the second half of the sentence moves into iambic foot um Woolf I think very consciously um replaces um the sentence in order to um make that um prediction error not just something that we rationalize but really something that you experience as a reader um on a quite embodied level so also the pacing of the prose the pacing of the language that we read um works as precision and guides readers attention um these features that Woolf chooses to modulate Precision here the different meters um of verse are of course Very Much tied to the period in which Orlando is at this point in the Noel Renaissance poetry um he's in the Renaissance where poetry is one of the dominant modes of um literary expression and that takes me to the third level of the probability design the Precision expectations so expectations about where we likely to find um good um Precision the kind of parameters um within which we direct our attention so um the setting in the Renaissance the fact that he writes um poetry um

goes together with the choice of the medium that here and both I think quite systematically develops that as she moves through the different periods so when we're in the Victorian period we read that sentences swelled adjectives multiplied lyrics became epics and little Trifles that have been sentence essays a sentence long we now encyclopedias in 10 or 20 volumes um you don't need to have read Dickens to know that Victorian novels are long um and Woolf actually here even in *Acts* she's not just saying that *Pros* um has a very long binded format long binded syntax in Victorian novels she actually performs that um here as well again drawing attention to the framework within which we're moving so in literary texts quite often um the genre or historically specific modes of writing provide Precision expectations for readers and indicate where it is worth paying attention to while reading so the probability design then brings together all three levels the level of the plot and the prediction errors the level of perspective and pacing and um precision and the level of different styles of writing different kinds of intertextual references on the level of precision expectations um it's an Artful configuration of words that provides a design sensory flow and reshapes readers generative models for the duration of reading and then Design sensory flow I think sits between the sort of everyday setting where of course we have a sensory flow but it's not um designed it's not ordered for us and cases of um for example *Dreaming* or mind wandering where we we're moving through the imagination um as Andy Clark says in *Serving Uncertainty* such systems of dreaming and mind wandering command generative model capable of reconstructing the sensory signal using knowledge of interacting forces in the world that process of reconstruction tuned and deployed in a presence of the sensory signal paves the way for processes of outright construction able to perform able to perform and develop in the absence of the usual sensory flow so in the case of reading we have an absence of the usual sensory flow but that doesn't have to be reconstructed completely on the basis um of what we already have what we already know instead it is reconstructed on the basis of this design sensory flow of the literary text that provides us with um prediction errors in the first place but then as I have shown um a whole set of different elements that also model um how we inference on the level of precision and precision expectation um what I talk about in terms of probability designs arguably can also be found in other kinds of artworks and um for example in the case of painting the composition would be a kind of probability design um and for that case the work of Gombrich um is very interesting because they talk about the ways in which the composition of a painting has the capacity to create particular aesthetic emotions um and that is tied to a faster than predicted rate of prediction error reduction meaning um that you see a painting and you're quite startled

it's not easy to make out what it is that you'll see but then because it is designed because the painting has a composition you can actually make sense of um what you're seeing in front of you at a much faster rate than you would predict at a much faster rate than what you would be able to do um in a real world setting and that creates a positive aesthetic emotion even if the painting perhaps depicts um something that would evoke quite negative emotions and I think something like that is certainly also true for literary probability designs that they have a way of um making you navigate um that probability design at a faster than reduction um but I would also say that there are quite significant differences between um paintings and literary texts just because of the way in which um they're made the way the things of which they consist in Reading literature um you usually have an engagement that extends over much longer periods of time than looking at a painting when you read a novel you spent probably about a week with it reading a couple of chapters here and a couple of chapters there so there is a way in which your everyday life and your reading process are folded into one another as you go through um reading that long form um fictional text um reading a novel also asks you to engage with visual stimulus but that visual stimulus is a lot poorer um than would you have in images it's just black letters on a white page so there are um quite important differences that I think is important uh to keep in mind both um literary texts as well as other art forms can create a feeling of presence um in readers and that's um based on work that I've done together with Federico and colleagues looking at Presence flow and narrative absorption where we distinguish between um motor intentions um that would be you handling the book and turning the pages proximal intentions um where you use language to act or give sense reactions of a character so you experience um embodied resonances around a character's movement for example while reading that generates um core presence and then there also distal intentions um where you use the probability design to make accurate predictions of what a character will think or feel so they develop a kind of strategy for getting better off predicting um what that character um will do um and that creates extended presence so it has a kind of second track or a meta track um to um the inferencing process and I think for the literary probability design that level of the extended presence is where most of the really interesting thing happens and I think that is to do with the way in which literary form or probability design can move Beyond mimesis can move Beyond just representing um something that is there I'll give you an example from a mid 18th century short story called the noble man where a young woman is escaping from the shadow of her ancestors and she does that by taking the portraits of the ancestors of the wall and using them as planks to run across the mold um around the shadow Julie jumps

lightly on the face of one of her ancestors as she leaves the shadow she jumps on one of the portraits um that face of her ancestors renss and snaps beneath her feet um there is of course here a highly embodied language that invites embodied residences and readers but at the same time um that design sensory flow also goes beyond the remit of real world sensory motor contingencies um I mean I'm I'm not sure it's possible to actually walk across uh canvas on water and not sink or I'm not sure how fast you need to run for that um exercise to work out that however is usually not a question um that we ask ourselves when we read that and that is I think because on the level of the probability design um all of that hangs together all of that has its logic um it is of course was very funny very ironic that she uses exactly the sort of um the elements indicating the authority of her ancestors the portraits um for her getaway so on the logic of the together all of that makes sense um and that enables um the literary text even though it draws very much on embodied resonances um to move away from um real world sensory motor contingencies and to reshape the inferences that are available to us moreover um Beyond these non-mimetic um moments liter texts very often invite what pulo and colleagues have called epistemic active inferences that is you want to make a change in the world in order to see um what provides better information and in the case of epistemic active influences you would do that in your head you would scope for information by manipulating um the environment mentally readers I think constantly do that you're constantly constructing alternative scenarios along the lines of oh if she hadn't done that or um that character really should have come in time um Etc um you constantly project alternative outcomes sometimes texts even give you prompts for that but these epistemic active differences that forward scoping while reading um is a fundamental part of the reading process and I think it is most prominent or most noticeable when you read the book again that you've read several years ago um because it then quite often turns into a very different um experience the kind of sense that you make of it the meaning that it has for you tends to be very um different just to give um a banal example reading anakaren before and after a divorce I expect would have a very different kind of um experience that goes with it and I think that has to do with the fact that yes we're reading the same text yes the plot events do not change um but the small uh inferencing processes that we run as readers um these epidemic active entrances are very much to do with the online present leing process and their fundament to the kind of meaning that we configure out of reading a novel and that changes of course between um different iterations of reading the text a second element um that I think is quite specific to um a literary probability design is the way in which novels allow you to explore inner states and of course the novel has

um traditionally been considered as the cultural technology for the exploration of IM States and I think there is a predictive processing story um to that um insight and it is to do with what Andy Clark in his uh 2013 BBS piece called the dark room problem um where he poses the problem of um well if we want to minimize prediction errors why do we not um just close the door shut lines and of the lights um one of the responses or the possible solutions that have been put forward for this dark room Problem by pulo and colleagues is to say well actually there is an inceptive and an exceptive aspect to this we keep our interception very stable so we don't want to have um a heart that beats very F we don't to be out of breath we don't want our tummies to be upset and if we keep that stable um um then we can be very adventurous exploring uh exception exploring um the outside world um and I would suggest that in the case of reading actually the opposite is the case we keep extra reception very stable um we don't want to be distracted we um focus on this very poor stimulus of just words on the page um keeping whatever attracts our extra reception um quite static and that then in turn allows us to be quite adventurous when it comes to exploring our inates the ways in which we respond to what we've read um very much of course emotional responses and in the work of Lisa Felman Garrett interception has of course been um tied within a predictive processing framework to emotional processes um but I think also things like um linking that to mind wandering Etc um would um fall within this exploration of in the states so the novel as a cultural technology exactly because it is um so stable when it comes to the stimulus allows for an exploration of interception and inner STS there's the second aspect to that um which is perhaps even more speculative and that is to do with the way in which which um embodied resonances might um be affected by the fact that we're reading that as letters on the page so in the study by shill and all um they were interested in the ways in which you can get around sensory attenuation so the ways in which you can lift um the attenuation of the sensory feedback that what you don't want to have while you're moving um they use the of um the escalation of force so if someone pushes at you and you asked to push back to the same degree you will usually push harder than the pushing um that comes against you exactly because you don't have a very good sense of how hard you're actually pushing um that kind of um Force escalation can be apparently um avoid it if you use a kind of interface that redirects um what it is you're touring and arguably words on the page are a similar kind of interface so using um an interface for pushing that redirects our movement in this case would limit sensory attenuation you would have a better sense of what you're actually doing in an embodied um sense and perhaps also embodied resonances of words on the page might have a similar effect now like I said this is

quite speculative but I think it would be interesting to see whether there is some kind of empirical reality to that and it would go together I think with this idea that literary texts as an exploration of inner states allow for more precise investigation of what's going on within us um then basic offline thinking would enable um the elements of reflection um can also be tied I think very much to the way in which prison is modulated in probability designs and here I suggest that literary texts have a way of constantly providing what I call Meta fictional nudges so nudges um that provide a meta perspective on the kind of immersive experience that you're having I take the term of the nudge from John Sutton's work on um Sports Performance where he talks about cricketers who use words for self-cueing um so that they can sculpt the ongoing flow of motion by saying things like battle or keep the battle something like that so you adjust what it is you're doing in the flow of the action and I think that's exactly what very often happens in literary texts I've given one example here from Francis Buffer's um the go Golden Hill where we're in the middle of a Fel Septimus disengaged with a WRA of Steel and lunch lower in SE Smith replied in C seek supreme but this is really useless and no more enables the reader to see the battle than if I shout numbers at you which indeed I appear to be doing so here you've got exactly that shift from a quite immersive description to these French numerals which of course indicate fencing decis uh fencing positions um first working as quite integrated quite subtle metafictional nudges and then being taken up by the narrator to a full-blown um meta fictional meta narrative reflection and that's something that happens quite commonly in literary text and I think invites readers exactly for that kind of uh shift of levels also the interconnectivity of text that I've talked about in terms of that third order level is something that I think allows L text to do some quite particular kind of cognitive work and here my example is a novel called de composition angara where the main character is a serial killer who works on the basis of PRS In Search of Lost Time and the rule that he gives himself is that he takes up a volume of pro um identifies the characters say Madame um counts when she appears in a given part of the book so she would be character number 37 who appears in that part of the book then he goes and sits in the park and counts the passes by number 37 if it is a woman he will go after them and kill them and then go home and uh open an electronic copy of Prest Noel and remove that character from it so here we have a case of a very strong emphasis on um that in textual Dimension that um Precision expectation Dimension and that is highlighted even on the level um of the words that she uses for example after the description of the kill rule um the narrator says deushi with happiness joyfully um you're going to bed which is a riff on the first sentence of BR novel where it's the B Earth so early um I used to go

to um so gar quite systematically uses not only the character repertoire of post um but also um the language to establish that interconnectivity um and I think that allows readers to reflect both on the rules and predictabilities of what's going on in that novel um but it also allows them to reflect on where they actually take their expectations from I mean is it the crime novel the serial killer narrative um is it the crime Society modern uh novel for which Cru um stands so as you're exploring Deton Library um this is something you can do um in the literary text exactly because it provides scaffolding for it because it ties it to different characters to titles of novels to ways of writing um Etc so that exploration of the mental library of different kind of precision expectations or sets of precision expectations and that is a term I take from PI the mental Library um is also I think something quite particularly suited for litery texts um because it allows a moment of stability and it makes these prediction are U sorry Precision expectations perceptible um by highlighting them and making them part of the story itself so to summarize briefly um what I think is quite particular about literally probability designs is the huge space they give to epistemic active inferences um it's the enabling of the exploration of inner states and I think that is in particular linked to this long form written text where you keep coming back um to the same fictional World while your everyday life um is going on um and thirdly that it allows you to attend to Precision expectations this third order level um of cognition which in an everyday setting in the ongoing flow of action um you cannot really attend to quite as easily so the long form written probability design of the novel um enables you to do all of that cognitive work um to an extent which is hardly imagin hardly imaginable in an everyday context um I think all of that is tied to form and probability designs um very much corresponds to a general notion of form and I take my understanding of form from fed Schiller's book on the aesthetic education of mankind he sees form as a means to provide freedom for the human mind and here of course the epistemic act of entrances this roaming of the mind um is one um element that's quite important but that I think is made possible by the scaffolding provided in the litery text exactly through that design sensory flow that keeps um these movements um discernable and thinkable that understanding of form as a space of Freedom as a space where you can watch yourself think um is quite different from how form is generally understood um certainly in literary studies Caroline lavine's recent book um forms which has um been very influential in literary studies says well there are social forms and there are also aesthetic forms so social forms for example would be the layout of a classroom and the aesthetic form of the novel um can pick that up and integrate it for example in a novel like um J A where we have a classroom setting um and she makes a very interesting argument

I think about the common durability between aesthetic and social forms but she also insists that forms as designed arrangements are deliberately crafted to impose order just like art objects um and I think is perhaps a little further than I'm willing to go because certainly probability designs provide patterns they provide scaffoldings um for us to give structure um to the experience of reading but I don't think they very strictly speaking impose order um for LaVine um the only way in which literally forms can create contingency can create a sense of that things things don't necessarily are the way um in which they appear to us now that there's always an alternative that um happens when you have different kinds of form collided but form in and of itself wouldn't give space to that kind of contingency exactly because it imposes order and I think the probability design idea allows you to rethink um the role of contingency in um literary forms uh and I've explored that in particular in my most recent project where I look at how authors create um probability designs how they write their novels um and there I suggest that contingency not only plays an enormous role in the creative process itself but it is actually inbuilt in literally form that also deploy form um to integrate contingency into that text to always face um for alternative ways of interpreting things making that available to readers so form of course encompasses contingency it is a contingency trap as I suggest but it's the kind of trap that you see at the bottom where the mouse will be caught um but it can still run around and it gets stabilized enough for you to observe the kind of things it's actually doing um form is not the kind of contingency trap that you can see at the top that just kills the mouse dead to give you an example of what one of these contingency traps um looks like um I'll briefly talk about italino castle of Ross Destinies where in the very process of narration you have this very alator um chance-based composition process where terot cards are placed on the table and the narrators develop their narrative from the card that they happen to draw and place on the table so on the level of narration we have a representation of contingency but much more interesting it also thematizes the way in which contingency is inbuilt in the form itself because in the first iteration of that narrative technique the Middle Point the middle spot of the tarot um cards remains empty and um this is work by Natalie burkeman um who suggests that this is a case of the cleam men um the swerve or deviation from a planned protocol um the swerve or Camen in terms of a probability design would be an underspecified element um that provokes further inferencing because the middle spot in um this ter tarot tapestry is kept empty you can always generate um new stories out of it and similarly literary texts always need to have some kind of flaw something that doesn't add quite up in order to um provide exactly the space um to explore contingency while reading so I

think in the very design of a probability design um there are very often these moments of a break or a flaw that enable these Freer um formal patterns of inferencing and meaning making with that in mind I want to come to the last part of my talk where I address the issue of how literature works as an uncertainty practice uh in the real world um and here I draw on an article that I published last year where I also draw on the piece by Miller and colleagues um where they talk about the value of uncertainty in the context of algorithm um and the way in which algorithms um make volatile environments more manageable um they talk about personalization algorithms so like when you order a book and you get five recommendations on the basis of what you've ordered before these kinds of mechanisms that make the world more predictable more um according to how you've behaved in the past but they also have a very interesting example of someone who programs an app for his phone um that randomizes his food deliveries so he never knows um what he gets so you can use these Technologies to go in both directions um and make sense of um the volatile environments that we live in they briefly touch on literature in that article and they say that it's a safe space or a safe kind of environment for exploration which in a way I think goes together with a lot of things that have been said about literature in a sort of larger um context where this is a space where your actions wouldn't have consequences where the killer from um gar novel doesn't come out of the novel to get you but of course you can also not save any of the characters in the novel so it is a space um of its own where you don't um need to fear consequences of actions so in that sense of course it is safe um but I think it is a lot more than just a curated space for Thought experiments um I think it is a lot more than that exactly because of the ways in which reading a literary text has a way of tying itself um into your everyday experience as you do that over multiple weeks because of the ways in which it draws attention to your reactions um to what you're reading so for example inos Madame or the person identified as Madame by the killer actually gets away and that goes very much against um what the whole novel was set up to do um and I think at that point you would perhaps ask yourself why perhaps you Shar the disappointment of the killer then you might ask yourself why you do that um or you don't do that and then you're wondering where the novel is going to go um so the way in which the liter text um is set up making things uncertain um or contingent is often also an invitation to reflect on how you react to that um yourself both with the exploration of inate but also in tandem with that metafictional nudges that I was talking about earlier on and I think that even applies for the question of um Precision expectations where the way in which this is set up in G to stay with that example from the moment that Madame Loop gets away we've got continually new

iterations of different character kills in different narrative modes um there even is an evocation of an ego shooter um where the killer based basically depopulates the entire world of In Search of Lost times uh In Search of Lost time and that variation of narrative speed um I think is a way of drawing attention to um the different regimes of attention that we go through in that novel The ego shooter going very fast very quickly and of course proves Noel because of the way in which he writes and how long the thing is um which grows very slowly so that would allow for example for reflecting on um the different kind of accelerations or attention regimes um that we operate in while reading that Noel herself actually in an interview draws attention to the way in which the experience of reading a book is by no means a safe environment she says it reorders the circulation of your affects it manipulates your desires your perceptions your efforts in ways that are not predictable so literature to conclude my presentation and its probability designs are actually quite strange think about Julie running across the portraits of her ancestors and not exactly safe because they've constantly forced you um to reflect on what's going on to reflect on how you think about yourself they have the potential to at least while the reading process is going on to reshape the generative model with this design sense Reflow and they make elements of cognition that are usually invisible reflection so they enable creative contingent ways to rethink yourself and the world and I think that's as much as we can ask of any kind of cultural technology thank you thank you very much Karen was really fascinating account of um how literature can be seen through the lens of predictive processing and an active inference and I think I would like to um because you dived really directly into how to do this how to use this framework to think about literature and how readers engage with the literature but maybe other question that can be interesting for a broader audience and this is a um so what do you think that this perspective is offering um that is new or that is more effective in terms of um how we can better explain what's going on in literature and when we engage with literature in comparison to um existing Frameworks or alternative uh theories that deal with the with similar um aspects of literature um that is an enormous question um I think one of the reasons why I've decided to spend quite a lot of time with predictive processing um to think about literature is that it allows you to look at multiple levels together so it allows you to think about the plot but also how that's connected to the character and how they move through the fictional world and then you can also make links to um yeah very particular kind of genre protocols for for example or intertextual links um so I think because of the way in which it works across different levels um and still the flexibility that I see um in the model um what I like about it in terms of thinking about

literation why I think is effective is that it actually allows you to think things together that traditionally in liter studies we don't think together so character and plot um are usually two different str of things that you're looking at yeah great answer and actually made me think about [Music] another maybe issue that it's uh very often uh debated or maybe not enough debated sometimes in in literary theories and is about the need or the possibility actually to have a comprehensive and Universal theory of of narrative of fiction of literature and by what what you've said and also this is my impression um in other Reflections that well it seems like that something like um thetitive processing framework and active difference similar to how they've been proposed in the cognitive Sciences can aim at being this um overarching theory for describing and explaining different aspects of narrative and so my question is do you think this desirable and it's possible and how this can be um can go together uh with the need uh that has been expressed multiple times in the humanities of having different perspectives and different um people with different theories that can actually bring us to a deeper knowledge of a phenomenon um again a very big question um I think with predictive processing you can ask very different questions of a text um and you're not tied to so if we're thinking about universals um that very often comes with a category or a set of categories so I don't know four different character types and then you've got everything in compass um so it very often comes with a top down um imposition of boxes so you've got the hero and the helper and just to take a very basic example of Pop's fairy tale studies the structuralist morphology um what I like about predictive processing is that you can do that bottom up um you don't have to look for a particular character category in a text um so you can with that account for um any number of narratives without um presupposing heart distinctions between types or structural boundaries instead you can trace how the text or the novel itself establishes um those categories or keeps them open I mean that I think that is a lot more common in in literary novels that the and um the hero um are not very easily to be distinguished um so I think one of the attractions is this bottom up element to break processing well that it allows you to do bottom up analysis that can all be analyzed with the same analytical vocabulary yeah thank you yeah I like the the the perspective and I'd like to to ask you last question before uh letting the the floor to to Bruno and I know that you in your work um and it's an example of how you uh presented this today you look at text from different epochs right not just contemporary material like to look at how this Theory can work with the with text from from the past so I always wondering how um this can work well uh this way of of looking at the text and thinking about probability so kind of cognitive process that readers are doing or um hypothetical readers are doing and how do you

in your experience in your analysis of the of several literary works that you've done how well do you think this is applicable to text from the past and uh or where do you see the limitations are um so I think it is very well applicable um in fact I got into predictive processing because one of the central categories of early modern literary theory is probability um and I was working on 17th 18th century literature looking at ways in which how they thought about what was happening when people read how that corresponds to current strands in cognitive science our philosophy of mind so um for probability I actually started um from the early modern period where it was very important that the text was probable um and went from there to see well actually predictive processing as a theoretical model Works quite well to understand how these um probability calculus um actually work so interestingly enough i i p the other way around um and certainly in the early modern period the way you thought about um many literary works not all of them was you know um not is it true or is it false but is it probable is it well enough designed does it have a good enough plot to hold together so that you believe it whether it's true or not um so I think predictive processing works very well um for I I mean one would have of course to put it to the test and one would have to apply it to Medieval texts or ancient texts as well um but certainly for the periods that I know which is mostly the early modern um period it it works quite well um it would be interesting to see what would be the the sort of limit cases where um the idea that you've got an ongoing um calculus that the trajectory of the narrative is Con updated um whether it wouldn't work I think that would be very interesting to see what kind of narrative that was thank you okay so I guess I'll just move on thank you uh for that lovely presentation and and for sharing your book it's been quite an interesting read and there really um changed a few things how I think a few things in my field um I just want to say that I come from a very different field um anthropology and archaeology but I think that a lot of what you say actually there's some parallels between the work that we've been doing um just to specify uh that worked mostly with the breakage of form and I I'll ask a few questions about this um uh particularly uh in terms of craft work so a lot of work I do is on Craft work and there's an interesting point you raise in in in the book about and and also in your presentation about how reading what what makes literature so exceptional in a sense what makes it different and I'm wondering if you find any parallels between which of the processes of probability design with uh with craft work with actual work with with materials um yeah I'll start with um so so you're thinking of the actual Act of making a pot for um yeah sure yeah um I think there are similarities certainly I think I I I have I have to admit I have no idea how you make a pot no worries um but I I assume it is also a more long form

engagement it's something that unfolds in the flow of time and um where the material that you're working with gives you different kinds of resistance um and you have different strategies of of working with it or working around that resistance so I think in in that way it's probably similar to the reading process that it is something that unfolds over time and where you get a sense of what is likely you know going to work um what kind of approach you're like to yeah it's is likely going to help you make sense of what it is that you're reading um so I think in that sense um there are a lot of similarities there um I think one of the differences is of course that reading requires you to work with language whereas the material um of the pot is not strictly speaking symbolic um I'm very happy to talk about the symbolism of a material form um but it's not uh first and foremost first and foremost it's matter that you shave um so I think there is a difference with um how you handle and manipulate that in your head um from what you do as a reader yeah I'm glad that you say that it's been something been debated a lot in our fields in terms of there's a turn around trying to make a looking at materials as as as if you were reading a text a lot of complication arose which you just synthesized very well I think but um I think I'll I'll shift to the the next Point really is is in terms of form I'm quite interesting this point because you seem to suggest that form is something emergent which I think is partly fueled by this U by the active inference framework um and if I understood correctly would be one would be in this sense um uh the design sensory flow yeah um and there's this tendency we have of trying to kind of petrify it maintain it make it durable um and uh but at the same time we're facing challenges right in the text from when so I would be interested in looking at if if there is form in the sense what would be the breakage of the form so what what would be something that challenges the form when when when when is reading or in terms of design um I think there are multiple ways in which you can do that um but before I start talking about that I I think I mean I would be interested in actually hearing your take on that because it it seems to me again as someone who's not very well informed when it comes to craft that if you do have a material object I mean you speak about fossil and I mean these are stable configurations M um that of course you can't perceive the whole pot at a time but you can get an outline of it so if you or if you look at the painting you know you can do a composition sketch um that puts together all the dominant lines and the outlines um so that you can see where the triangles are on the center of gravity or um the the the vanishing point Etc so it seems to me that in the case of visual art and and also material objects there is a way in which the form is much more immediately perceptible whereas um in a literary text and especially in a long form text like the Noel um it's really only something that yeah it's it's

somewhere there in the text but I think as soon as as you try to draw a diagram uh as soon as you try to you know do these plot lines or whatever people have attempted um it sort of doesn't quite capture it I think you do have an intuitive sense of um what kind of shape the whole thing has and I think there are various strategies when you start a Noel um of projecting what kind of ending it's likely going to have um evoking General protocols for example is establishing a setting that will come up again at the end Etc so there ways of signaling formal coherence in a text um but I would be hardpressed to imagine how that would have the same kind of perceptual Unity I think that I'm looking at AAS for example has um still I spoke of course about um cracks and flaw in um the probability design um and I think those are um exactly these moments that are heavily um underdetermined um that like the the character getting away in in the Cru killer novel when everything has been set up to say well this is not going to happen this is completely impossible and it it's also never explained properly um one example that immediately I think comes to mind or another example would be sort of what happens between chapters when characters at the end of the chapter Clos the door and then the new morning starts the next chapter and you you sort of know what happened in the night but it's not made explicit so I think there are various points um in the narrative or in the novel um where you can position those cracks and and those flaws um and I think some of them are to do with the very material layout like chapter endings um but mostly I think it will be on the level of how the narrative is constructed does that answer your question yeah I think it's true is it must more of a it's much more fluid this idea of of of of form really so it's hard to disentangle the this what would be the exact let's say not the exact opposite but more or less the case about how how form is really dissolved if to phrase it that way um I think there's a point I I want to ask you about um in terms of because you mentioned like uh plot events or these things that that did not happen a lot in and maybe I'm referring to exclusively on on your book in that case all about the this narrated and um and I'm Wonder wondering you you raised a really interesting quote that um I think I have it written here I can say directly yeah the unfulfilled possibilities or to put it more dramatically the hypothesis that die with each transformation in a plot event stay with the probability design and I actually think this is quite a crucial remark in that sense to understand how these let's say interruptions that are flowing through the the text actually come to shape the the this form as you as you're describing um so my question is are they would it be is there a selective process into how some hypothesis might die out why others stay with the POs sorry probity design is that um maybe so bit confusing the question but um I think I understand it um so I think that is very much to do

with the individual reader um and how you move through the text so if we think of a hypothesis that dies um I don't know that an now St with a husband you know which is possible at multiple points in the book um it is a very dead um hypothesis though but of course you could read the novel and at every point you know imagine how you know her life could have been different it may even have been better um I think that depends on your take on Roni um but I think it is very much to do with um exactly the kind of epistemic active inferences that become dominant across the reading process and it's very much to do I think also with the kind of emotional investment that might have as a reader because you feel sympathy with a particular character or because you think oh this character is a lot more interesting than I actually thought um often when you read a novel *The Second Time Around* and you start noticing the things that you haven't noticed um the first time around um so I think I don't think there is um I don't think it's predictable I think that is exactly which of these hypothesis that stay with the probability design even though they don't get realized I think that very much depends on what happens um during the reading process what for whatever reason is attractive for a reader to carry with him or her through the text okay I see your point um in terms of this rereading I think maybe I'm passing too much time so I'll leave this as the last question before now um in terms of the rereading uh process um and these ideas that these processes that are adding uncertainty let's say to the to the narrative um so would would that be reconfigured in terms of these these hypothesis with the rereading um are some of these taken or are they some of them just left to die eventually because of this constant reevaluation or reiteration of of the of of looking through the text and for the reader I mean yeah yeah I think most of the you know possible Alternatives will never cross your mind um because there are just too many of them um so I think the the one if you do carry an alternative hypothesis with you um it will be one that for one reason or another you have an emotional investment and all you find particularly interesting at given point um so I think it it leaves potentially an infinite range of making epistemic active influences but you won't have the time or the interest or the energy to make all of them um I think it will be a very few of those okay see Point um yeah sorry I said it was the last one so I'll keep my promise yeah maybe I can add one small things it's a comment on your previous question and to C answer because we touched upon this in the article me and K have written together and then you mentioned today this distinction between core presence and um extensive presence right so the difference between the proximal intention and the distant intention so you can make predictions and and then accept the errors coming through out of them as long as they don't disrupt too much the the goal that you have

the the the distant intentions that you have and they are driven by situational factors of by Jean convention let's say that you're reading a detective novel if you keep making predictions about who the murderer is and the murder gets revealed on page three that would be a to an abrupt disruption for you to accept that and then wanting to keep engaging with that kind of narrative and and and so you you ask if there's a trade-off or some uh things that um are not included in the probability design or get discarded in the writing process because they're not they don't work well in the overall design so maybe this is an example of something that could in some cases if you're really not if you're writing something that is subverting the Jean convention and not really um putting the emphasis on who's the um the murder but wanting to play with the the Gen of the detective story but this this could work but in most cases for most readers that want a detective novel um that wouldn't work I see um that's is quite interesting because uh what I see is all these what I find interesting because for us when we're working with craft work on certainty seems to be of the external world you know it comes in it everything's messy um the things Decay they wither they break down so you need to adjust whereas here find it interesting because a lot of this uh uncertainty comes from the actual design of the of the of the text as as as it's been uh described and so in that sense um is there an element to this to this design in the writing that one can one can try to see these these um let's say these hypotheses that uh that were never written do you find traces of these in the design for example when when when the text is designed do you find traces of these doubts let's say of the of the writer when when they were do you know what I mean something that might be external to that was intended but is can can one highlight these in the text about about uh the moments of doubt the author has when they're writing it the reason why I'm saying that is that is there uncertainty that is created inadvertently when it's when the text is being written that's that's my point I don't know if it's yeah I think um that certainly happens because I mean I talked about how reading is a long-term process but of course writing a novel is an even longer process um where if you look in into manuscripts you can see authors you know changing their minds um and that will leave traces um as they keep copying and editing um what they've written um and again with a text that is as large as a novel is I mean all of these are of course meticulously edited but I think there is space for you know the mess as you call it um to come in from the different manuscript um stages um sometimes that even continues after publication um one of the early modern authors I've worked on Samuel Richardson um he wrote a novel called Clarissa where one of the protagonists is an aristocrat and he thought he had written him in a very unattractive way and nobody would like him and of course all the

readers or many readers thought you know he was the real hero of this piece um so there must have something gotten in there um that he didn't so it's not a question of um you have an intention and then you write that there's always stuff that gets in there I don't know drawing on cultural Protocols of Rish Aristocrats that have attraction um without meaning um to do so and then of course I mean Richardson then spend you know various additions trying to edit out those elements that he didn't want in there in the first place so I think that that is a very common um phenomenon actually that you've got um a lot of mess um even in in a very well designed design okay see I can relate to that it's it's a very interesting question if I have time to also add the small thing Daniel is it fine yeah no because well some may even argue that so the uncertainty but not in the sense of of the mess right of the but in the sense already of of how uncertainty is defined by productive processing might be um the core uh quality of uh of literature that lasts long and actually so Ken presented the work BYO calino and he has written an essay um um I think the name was is what is a classic and and his definition of what is a classic is precisely this text that have something which is left open so a level of uncertainty and it can pertain to many of the different aspects that the carent touched upon that is is uncertain because of the overall design of of the text and and how it works and it's open also for future generations and that's why it becomes a classic right because this uncertainty gets transmitted and and the text gets transmitted because of this um leing uncertainty that appeals to to readers also of the future that that calino couldn't uh think of or any any author of Classics couldn't think of it's quite quite fascinating because something that stems to mind this is how would these different scenarios of uncertainty I mentioned uh something that you Ken used a an example in her book about these things that do not happen really so how how would those be scaffolded in the text you are they is that um for the reader to be able to follow through on these details does it make sense are you saying it it should be there there should be textual signals that you can point to yeah yeah so how would it be okay I'll give you an example maybe I'm thinking too much about craft so what we see normally is that when when we set out to learn about let's say you mentioned Potter so I'll throw in Potter um set out to make a part and the last thing in your mind you know that you encounter problems the most thing most of the things you want to learn about is the techniques to make the object right but the last thing in your mind is what will all the that what you will effectively learn eventually are actually all the faults that will come out throughout the whole process right but you don't intend on learning this and it's not explicitly taught many of the cases okay so a lot of this uncertainty comes from the exposure to these contingent situations yeah um and the problem

here is because it's not explicitly taught there are ways in which regardless if it's not explicitly taught what I mean is there are ways in which somehow it's there's a sort of scaffolded learning in the process of something that's not explicitly stated and I'm wondering so for example you would say the first thing you learn when the fault appears is um is affecting what is the fault you know what what consequences would this have you know then the next stage you learn okay um what do you do in this situation and then at the last stage you would say you learn to anticipate it right so you learn to anticipate this uncertainty yeah and so I'm wondering if it's a similar and maybe I'm thinking too much about craft but I'm wondering is a similar process with learning all these um let's say situations where uncertainty is being fed into the narrative to to us to uh get the reader engaged let's say with the with the story so does that make sense a little bit better maybe I think it does though I'm beginning to wonder whether that might not be a point where craft is actually quite different from Reading because I think in the craft you want your strategy to anticipate um those flaws in in order to not have them have whereas I think when you're reading a literary text you actually do want that kind of opening um you want to be given that kind of Freedom um so I think you have a similar process of developing exactly that um the capacity to anticipate you know if if the Killer is revealed on page three um if it's a standard crime novel um you don't expect much more but you also know that if someone I don't know like go chrisy does that so maybe there is then something else coming that Pro provides a new way of integrating that information in a probability design that is interesting in a different way because you know who the Killer is but maybe there's a new question that you hadn't even thought about um so I think there is similar development of let's call it expertise um as you work with the material in that case of reading a book um but I think you don't this anticipation is not to prevent um the floor from coming about it's we need to see okay so now I anticipate the floor and I want to see what it yields I want to see um what the consequences are so I think there is a difference there um and it may be to do yeah I'm not entirely sure um what element that would be um based on but perhaps it is to do with um letters being I mean they they don't need the kind of structural Integrity that a pot needs to hold together so maybe there is something on the very materiality of what we're working with that allows for um that kind looseness MH I get your point that's certainly true yeah sorry yeah towards a a pause in the chapter a chapter break here thank you all for the work I think it the the book and the other writings have opened up many avenues for people to explore and I'll ask one question which is art has always been cognitive how though has this cognitive turn in explicitly exploring the generation recognition and interpretation and

Analysis of art how is that explicitly cognitive turn re-entering into art itself um that is a very good question um I don't actually know um I mean there are obviously writers who have a great interest in various kinds of cognitive theories um so who read that because they're also very self-reflective and interested in you know how this works and so arguably there are writers that take up um cognitive science Neuroscience um Etc I'm not familiar with anyone who's taken up predictive processing though i' be of course very interested in that um so if you're talking specifically about the the sort of creation then I think it depends very much on what writer read does that answer your question yeah it's uh I mean it's an opening to see how this all happens but like the craft maker was already doing what they were doing what does an explicitly cognitive lens and a unifying cognitive scientific approach to memory anticipation the layers of probability designs that you articulate how does explicit knowledge of these in contrast with an implicit tacit craft Artisan approach to generating I mean icting is still in play even if we were to take that into a sort of forensic literary cognitive science approach so certain things are in common and yet now increasingly aspects of the cognitive elements maybe in the future people are reading with augmentation and there's new accessory possibilities for interoception while you're reading or there's generated texts not just recommended or Word of Mouth dissemination but instantaneous generation of designed probability design the these are these are shaping the structure of the form in the media yeah I think what you rais about the implicit and the explicit is very interesting because on the one hand of course it is an implicit process I mean you you go and you do it um on the other hand I do think that the this meta dimension that I highlighted with the nudges um it's quite important especially for the case of literature that there is a constant element of um adjusting taking a second order perspective um so you could very well use um say an analysis of a Noel appropriative processing where you actually identify and tag the elements um in order to think about what it is that you've been either reading or writing so I can see very well that um that kind of analytical language can provide um nudges or metap fictional nudges um for readers or writers so I think there is a place to think about predictive processing not only in terms of the theory that may explain um what is going on and how how it hangs together but that could be used practically um as well perhaps the panelist and then Karin any closing thoughts or how you take this forward perhaps Bruno then Federico if you have any comments well I think um I'll be brief but uh I think there's lot of for me personally is got me thinking a lot um how exactly the sequences of events are structured in in my line of work and uh how to look at these things a little bit more fluidly um with all these interruptions maybe I think we tend to be

a little bit more schematized with with with the way we engage with materials and so I think this is a way of actually understanding the process as as it's as it's unfolding really um and uh yes I think the the second part has to do a lot with this idea of uncertainty and and contingencies which are seldom used in in in our field so I think it's been really inight in that in that regards to look at um how this is actually an inbuilt part of the design process and and uh people come with very different ways in so as to how to respond to these different um elements sorry that's very very interesting perspective in see how um looking at archaeology and crafts can can have commonalities with the with and I I think um what I would like to see done more and also this what we've been doing when when writing this article together and thinking about okay this is it works with with narrative and and literature but then can we use this framework to see also what is specific about literature as as a text in comparison to to other media right because we live now in a society and in an era where literature liter text are not the primary medium through which people experience fictional words and narrative so this I think because of how it can work well across different mediated experiences I think the active INF framework can be a very um good way to define what is specific to certain medium or a certain genre even if you want as a the specific type of using a medium in in a certain way um in comparison to how can tell stories uh through a different media so I think yeah I think this would the the the line I would like to the line of research I would like to read more about thank you I think it's really interesting how we're coming back to the question of the medium um which has been so much discussed for media that are not literature um which has been discussed for film and virtual reality Video Games Etc but not as much for literature the fact that this is a written text that we spend so yeah this much time with um which is it has an effect on it and which is different from the craft example that um Bruno brought um so I think from from our discussion now that is something I want to think more about um the the the fact that this is a medium in and of itself and that exactly specificity that filo highlighted is very much rooted in meeting cool the kinds of things that happen on the live stream medium thank you though all this is very awesome I hope people enjoy it too so till next time thank you very much thank you e e e